

TITLE PAGE

SecureMailScope

AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications

Problem Statement ID – SIH26159

Problem Statement Title – SecureMailScope: AI-Assisted Cryptographic Security Posture
Assessment for Secure Email Communications

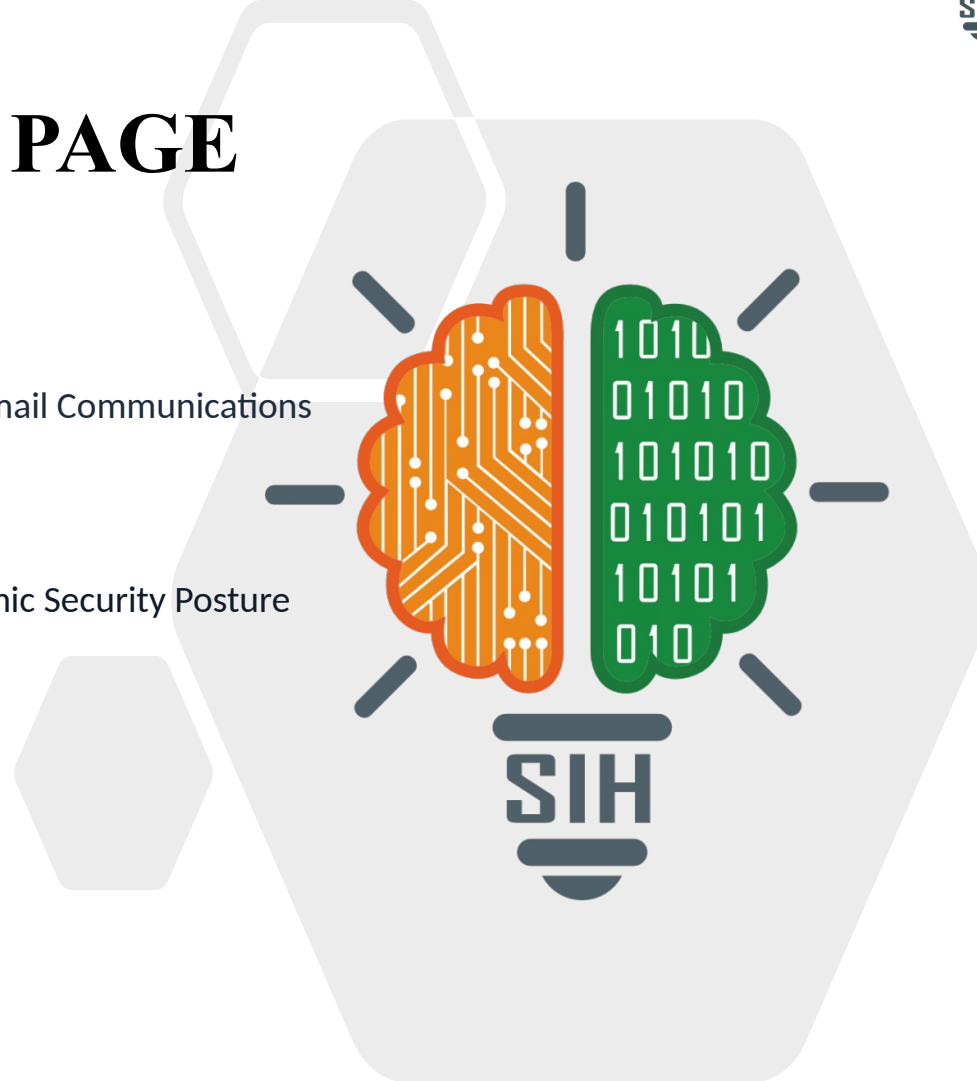
Organisation – National Technical Research Organisation (NTRO)

Theme – Blockchain & Cybersecurity

PS Category – Software

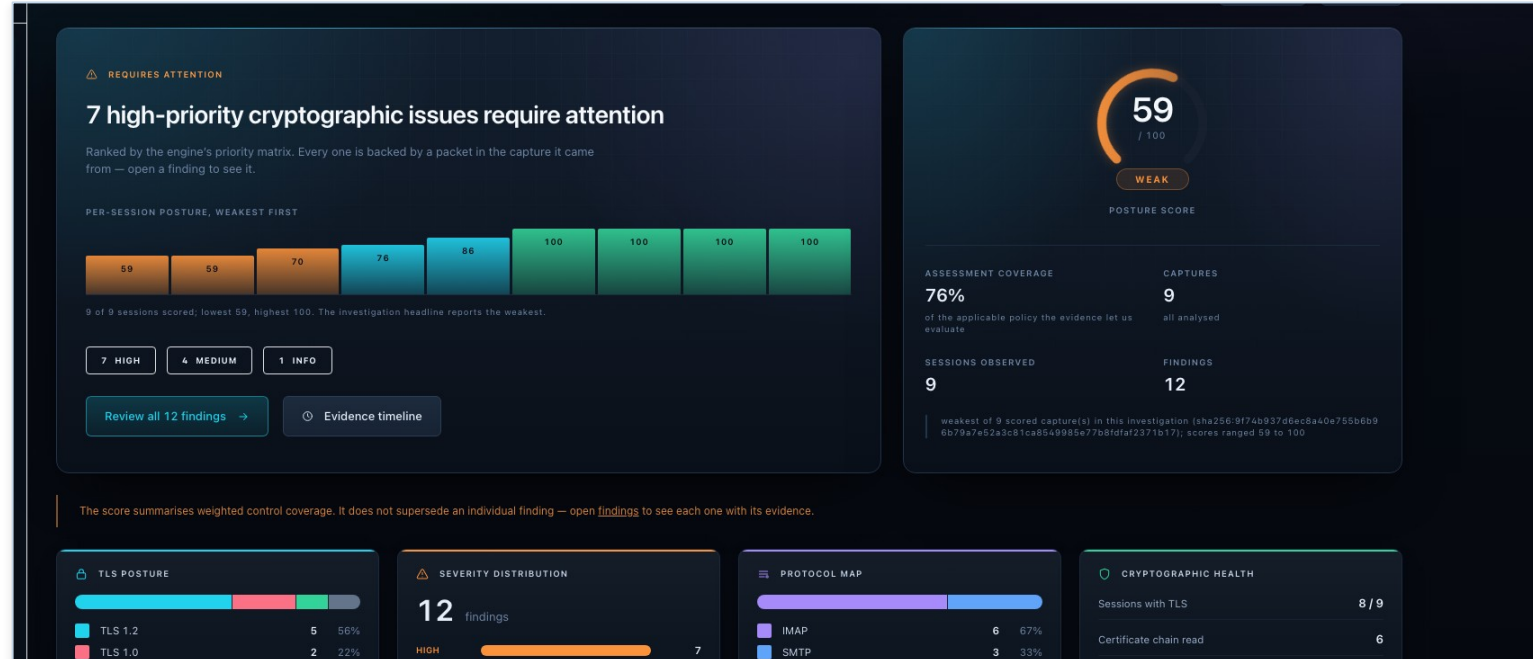
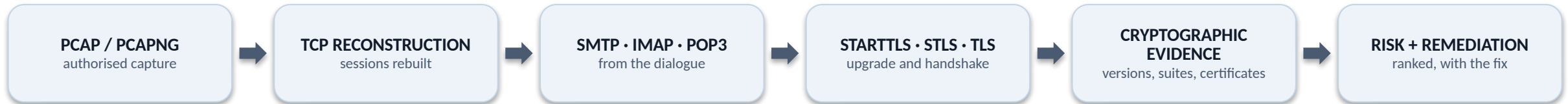
Team ID – 146876

Team Name (Registered on portal) – Zero-Day



IDEA TITLE

SecureMailScope turns passive email packet captures into evidence-backed cryptographic investigations.



The investigation opens on its conclusion. Actual product output.

PASSIVE BY DESIGN

Existing captures only. No live probe, no connection back to the mail server, nothing leaves the machine.

EVIDENCE-FIRST

Every finding traces back to the packet-level observations that establish it — number, timestamp and stream offset.

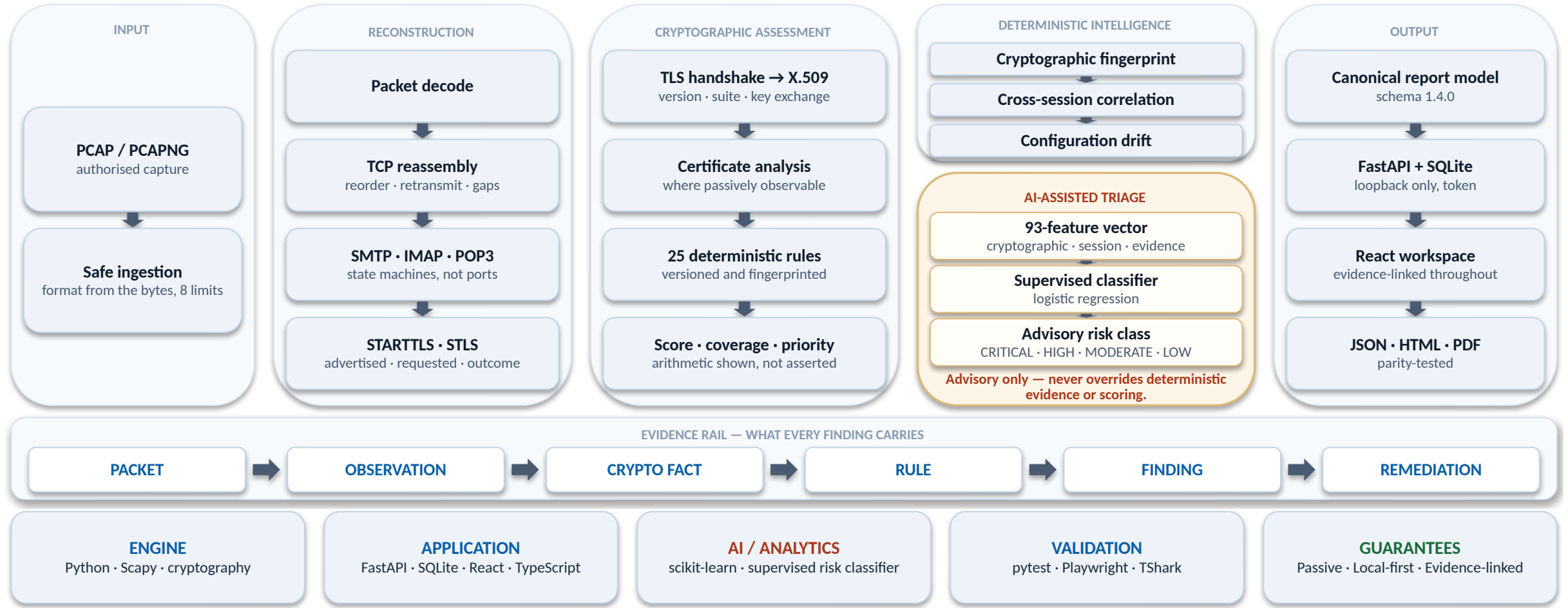
AI + CROSS-CAPTURE INTELLIGENCE

ML-assisted risk triage alongside cryptographic fingerprints, configuration drift and correlation.

TECHNICAL APPROACH

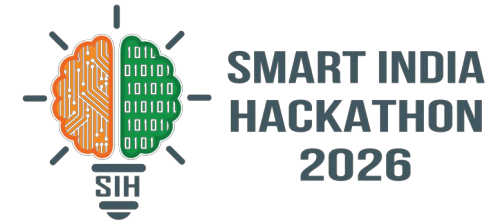
Zero-Day

Each stage consumes the previous stage's evidence and preserves packet-level provenance.



The classifier is trained on 313 synthetic sessions and is NOT VALIDATED for real-world risk; deterministic packet evidence remains authoritative throughout.

FEASIBILITY AND VIABILITY



Feasible to build, feasible to operate, feasible to deploy and feasible to maintain — each with its own evidence.

1,357

backend tests pass

1,367

with the TShark cross-check

89

frontend tests

5

real-backend browser E2E

158

files ruff + mypy clean

0

known advisories at audit

TECHNICALLY FEASIBLE

The pipeline runs end to end today: PCAP → reconstruction → cryptographic assessment → findings → reports.

Every figure in the strip above was produced by running the gate, not by asserting it.

OPERATIONALLY FEASIBLE

No agent installation · no mail-server modification · no credentials required · no active probing.

Works from the authorised PCAP / PCAPNG an organisation already collects.

DEPLOYMENT VIABLE

Local-first: FastAPI + React + SQLite. Core analysis works offline.

Captured traffic never needs to be uploaded to a cloud service; it runs as a self-contained investigation workstation.

MAINTAINABLE / EXTENSIBLE

Versioned policy rules · modular protocol and TLS pipeline · canonical report model · independent TShark cross-check.

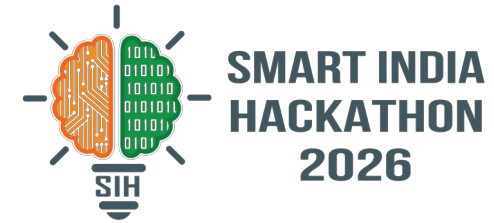
New deterministic rules can be added without redesigning the capture pipeline.

KNOWN BOUNDARIES

TLS 1.3 may encrypt certificate evidence a passive capture cannot recover; it is reported NOT AVAILABLE with the reason, never guessed.

The AI classifier remains advisory until real-world validation; it never overrides a deterministic finding or the score.

IMPACT AND BENEFITS



What changes for the organisation that adopts it — and for the people who have to act on the result.

REDUCES MANUAL ANALYSIS

Turns captures into structured cryptographic findings instead of an analyst reading every TLS negotiation by hand.

NO PRODUCTION EXPOSURE

Assesses posture without modifying the mail server, deploying agents, scanning or using credentials.

MAKES FINDINGS DEFENSIBLE

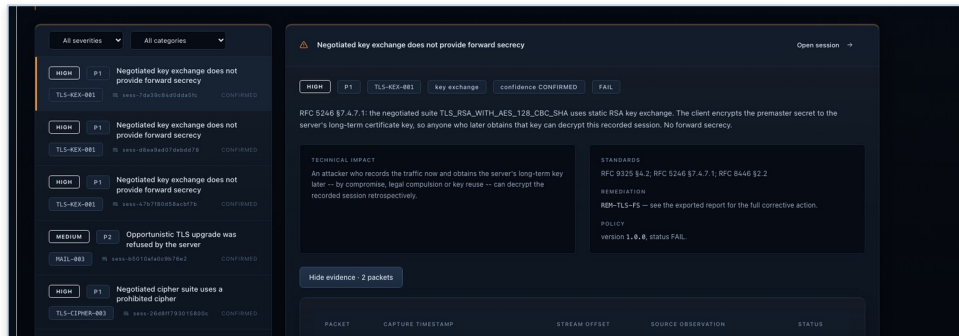
Each conclusion links to capture, session, packet, timestamp, rule and remediation — independently verifiable.

PRIORITISES REMEDIATION

Severity, confidence, priority and a named fix, rather than raw protocol data.

DETECTS REGRESSION

Cross-capture drift reveals changes in observed cryptographic posture between captures of the same service.



DEFENSIBLE — the finding, its rule and the packets it was evaluated against.

REGRESSION OVER TIME

negotiated version · OBSERVED_CHANGE

EARLIER
TLS 1.2



LATER
TLS 1.0

The same observed service, two captures. The change is recorded as OBSERVED_CHANGE, never inferred.

KEEPS SENSITIVE TRAFFIC LOCAL

Captured enterprise and government email metadata does not need to be sent to an external AI or cloud service for core analysis. The engine opens no socket. The only traffic is the browser talking to 127.0.0.1.

SOC / SECURITY ANALYSTS

INCIDENT RESPONSE

MAIL ADMINISTRATORS

AUDIT / ASSURANCE

GOVERNMENT / ENTERPRISE
SECURITY

Benefits demonstrated on controlled synthetic captures; no production deployment and no measured time-saving claim.

The standards the rules cite, and the pipeline every claim in this deck is checked by.

CORE TLS / PKI

RFC 8446 — TLS 1.3
RFC 9325 — secure use of TLS
RFC 5280 — X.509 and CRL profile
NIST SP 800-52 Rev. 2

EMAIL SECURITY

RFC 5321 — SMTP
RFC 3207 — STARTTLS
RFC 9051 — IMAP 4rev2
RFC 1939 / RFC 2595 — POP3 and TLS

VALIDATION / TOOLING

Wireshark / TShark
Scapy
cryptography
pytest · Playwright

HOW EVERY CLAIM IS CHECKED

CONTROLLED FIXTURES
one fixed seed, identical
bytes everywhere

**HAND-DERIVED
EXPECTATIONS**
written by hand,
committed, not read back

ENGINE UNDER TEST
1,357 tests; captures stay
out of git

**INDEPENDENT TSHARK
CROSS-CHECK**
against Wireshark's own
dissector

**REAL-BACKEND
PLAYWRIGHT**
five browser specs,
including a restart

REPORT PARITY
JSON, HTML and PDF
asserted to agree

PROJECT

GitHub repository <https://github.com/sgtsujith141-wg/securemailscope>
Demo video included in submission package.



SCAN FOR THE REPOSITORY

Public — source, tests and submission package.